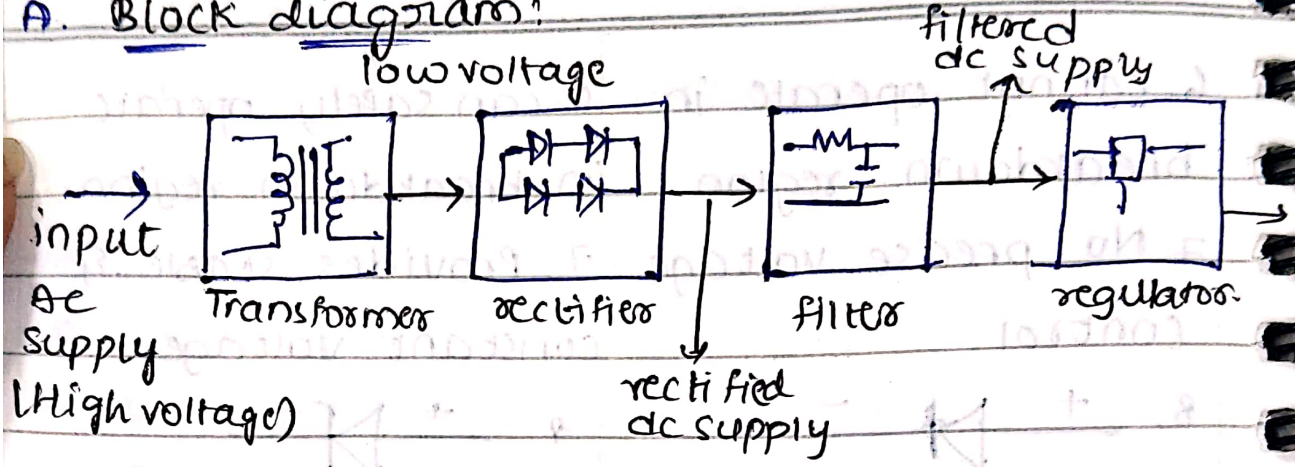
UNIT- 5

1. Explain the block diagram of regulated power supply.

A. Block diagram:



A regulated power supply converts AC mains voltage into a stable & constant DC voltage irrespective of changes in load or input voltage.

The block diagram mainly contains:

* Transformer
* Rectifier
* filter
* Voltage regulator
* load.

→ Transformer:

* Steps AC mains voltage up or down to a required AC level.
* Usually a step-down transformer is used
* Provides isolation from the mains for safety.

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→ Rectifier:

* Converts AC to pulsating DC.
* It is 3 types. Half wave, full wave & bridge
* Bridge rectifier is most commonly used.

→ Filter:

* Removes the AC ripples present in the rectified output.
* Usually uses a capacitor filter or LC/RC filters.

→ Voltage regulator:

* Provides a constant DC output even if:
  → input voltage changes
  → load current changes
* It has many types like, zener-diode regulator, series regulator & IC regulators.

→ load:

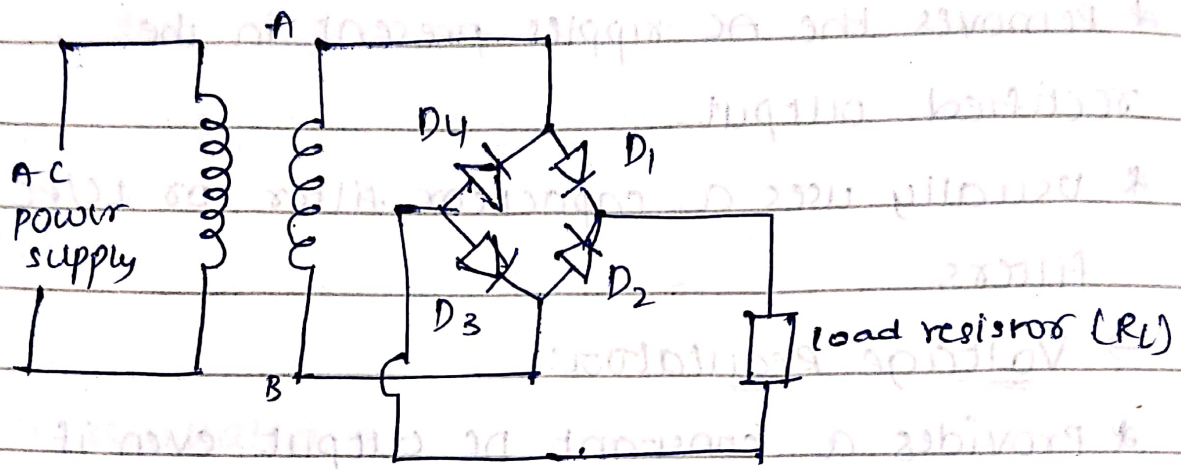
* The device or circuit which uses the regulated DC supply.

2. Explain the operation of full wave rectifier & draw wave forms & derive the expression for output voltage & current.

* A fullwave rectifier converts both the positive & negative halves of an AC signal into pulsating DC

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* It uses either two diodes or four diodes to make current flow in the same direction through the load.
* It is more efficient than a half-wave rectifier because it utilizes the AC cycle.



→ Working:

(i) Positive half cycle:

* During positive half cycle, the upper end of AC supply becomes positive & the lower end becomes negative.
* Diodes D_1 & D_2 conduct & D_3, D_4 will not conduct.
* Current flows through load resistance in one fixed direction.

(ii) Negative Half cycle:

* During the negative half cycle, the upper end becomes negative & lower end becomes positive.
* Diodes D_3, D_4 conduct & D_1, D_2 willnot conduct

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* Current flows through the load resistor in the same direction.

Thus, both the half cycles are used giving full wave rectification.

$$V_{dc} = \frac{1}{\pi}\int_0^{\pi} V_m \sin\omega t \, d(\omega t)$$

$$= \frac{V_m}{\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$= \frac{2V_m}{\pi}$$

$$I_{dc} = \frac{V_{dc}}{R_L} \Rightarrow \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi}$$

$$\boxed{I_{rms} = \frac{I_m}{\sqrt{2}}}$$

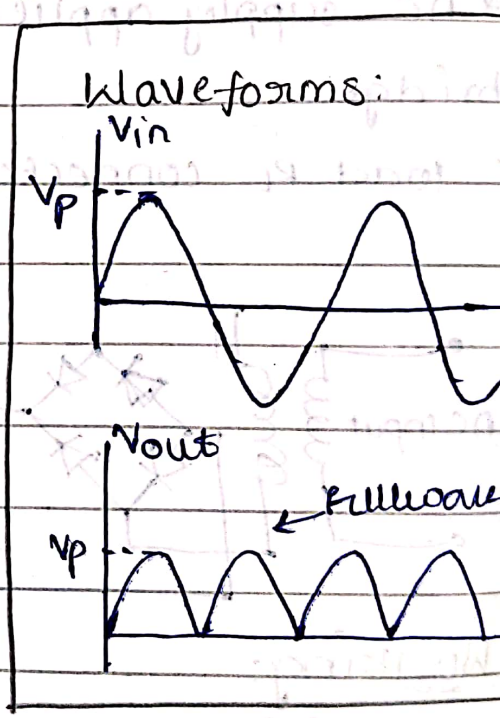
Full wave rectification efficiency $\eta = \frac{P_{dc}}{P_{ac}}$

$$= \frac{(2I_m/\pi)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 (r_f + R_L)}$$

$$= \frac{8}{\pi^2} \times \frac{R_L}{(r_f + R_L)}$$

$$= \frac{0.812 R_L}{r_f + R_L} = 81.2\%$$

Waveforms:



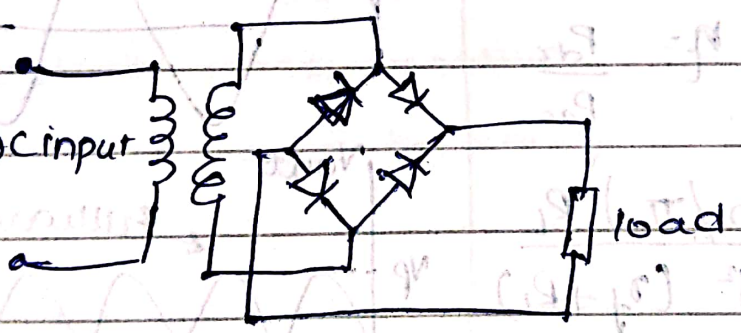
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3. Explain the operation of bridge rectifier & draw the wave forms & derive the expression for output voltage & current.

A Bridge rectifier is a type of full-wave rectifier that uses 4 diodes arranged in a bridge to convert both halves of an AC input into a pulsating DC output, without using a center-tapped transformer.

Construction:

* Four diodes D_1, D_2, D_3, D_4 connected in bridge form.
* AC supply applied to one diagonal of the bridge.
* load R_L connected to the other diagonal.



Working:

(i) positive half cycle

* D_1 & D_2 are forward biased, D_3 & D_4 are reverse biased.
* load current flows in one direction.
* current passes from source, D_1, load & D_2.

(ii) Negative Half cycle:

* D_3 & D_4 are forward biased, D_1 & D_2 are reverse biased.
* load flows in the same direction as before

Thus, both the half cycles contribute to O/p. output frequency is twice the input frequency

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4) what is Adder? Explain half adder & full adder and draw corresponding truth table & logic diagram.

Adder:

An Adder is a combinational logic circuit used to perform addition of binary numbers It adds two or more binary digits and produces a sum and a carry as output.

Adders are the basic building blocks of digital Cmputers, Calculators and ALUs.

Half Adder

A half Adder is Cmbinational circuit that adds two single-bit binary numbers and produces:

- Sum (S)
- Carry (C)

It does not Cnsider any Cary From Previous addition.

→ Inputs :- A, B

Outputs :- Sum, Carry

Truth table

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

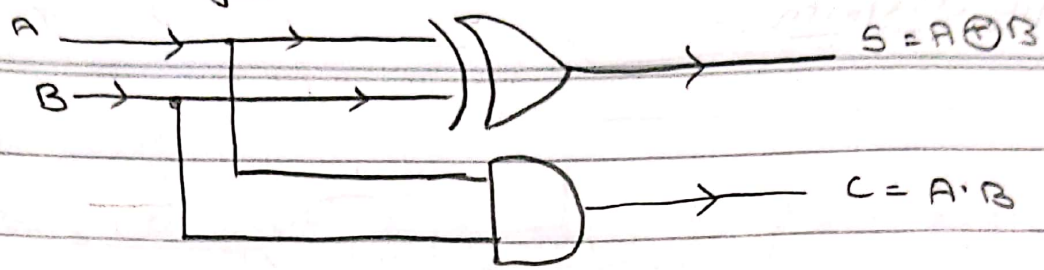
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logic equations

$S = A \oplus B$

$C = A \cdot B$

logic diagram



Full Adder

A full adder is a combinational logic circuit that adds three single-bit binary numbers.

→ A → B → Cin

It produces:

- Sum(S)
- Carry-out (Cout)

-> It is used for multi-bit binary addition.

-> Inputs : A, B, Cin

outputs; sum(S), Carry(Cout)

Truth table

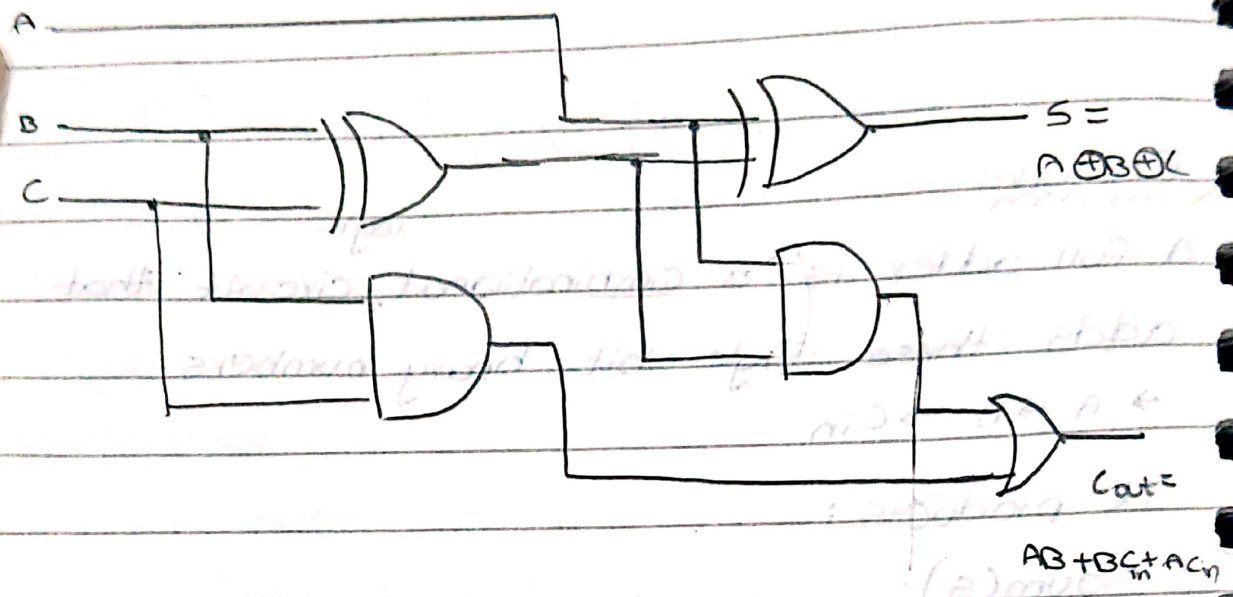
A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

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logic equations

- $S = A \oplus B \oplus C_{in}$
- $C_{out} = (A \cdot B) + (B \cdot C_{in}) + (A \cdot C_{in})$

logic diagram :-



5) Explain different types of logic gates with truth table.

logic Gates are the basic building blocks of digital circuits. They perform one ~~(or) more~~ bin logical operations on one (or) more binary Inputs and produce a single binary Outputput.

Commonly used logic gates are:

1) AND Gate:

The AND gate gives output 1 only when all inputs are 1.



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Truth table

A	B	A·B
1	0	0
0	1	0
0	0	0
1	1	1

2) OR Gate

The OR gate gives output 1 when any one of the input is 1.

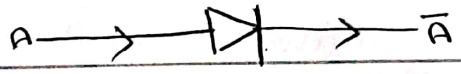


Truth table

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

3) NOT Gate

The NOT gate (inverter) gives the complement of the input.



A	$\bar{A}$
0	1
1	0

4) NAND gate

The NAND gate is inverse of AND gate O/P is '0' when all inputs are '1'



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Truth table

A	B	$\overline{A \cdot B}$
1	0	1
0	1	1
1	1	0
0	0	1

5) NOR gate

The NOR gate inverse of OR gate. Output is 1 only when all inputs are 0.

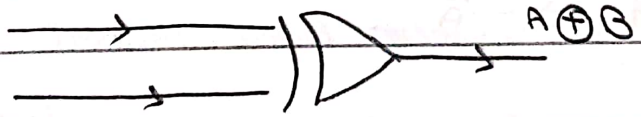


Truth table

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

6) X-OR gate

It gives 1 when both inputs are different.



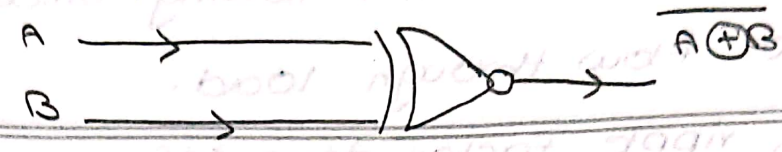
Truth table:

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

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7) X-NOR Gate

"The output is true if both inputs are same".



Truth table

A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

9) Explain L-filter & C-filter and draw the corresponding wave forms.

A filters is an electronic circuit used to remove AC ripples from the rectifier output and obtain a smooth DC voltage. Filters are connected after the rectifier and before the load.

Common filters:

1) L filter (inductor filter)
2) C-filter (Capacitor filter)

1) L-filter (Inductor filter)

An L-filter is uses an inductor (L) connected in series with load to reduce ripple content in the rectified output.

working:

→ The inductor has low resistance for DC and high ~~resis~~ reactance for AC.

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→ when ripple current increases, the inductor opposes change in current.

→ AC ripple is reduced, allowing smooth DC to flow through load.

→ Hence, ripple factor decreases.

Advantages:

1) Simple construction
2) Good voltage regulation
3) Suitable for high current applications.

Disadvantages:

1) Large and heavy.
2) less effective at low load current.
3) Costly

Applications:

1) power supplies.
2) High current DC circuits.

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C-Filter:

A C-Filter uses a Capacitor (C) Connected in parallel with load to smoothen the rectified output.

Working:-

1) During peak voltage, the Capacitor charges quickly.
2) When rectified voltage falls, Capacitor discharges slowly through the load
3) This fills the gaps blw peaks, reducing ripple voltage.
4) output becomes smooth DC.

Advantages:

1) Simple and economical.
2) High ripple reduction.
3) Compact size.

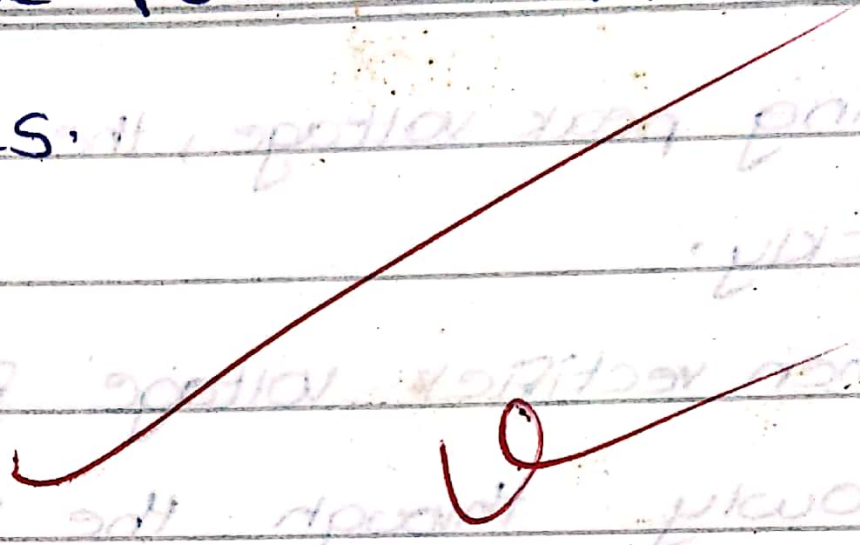
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Disadvantages:

1) Poor voltage regulation.
2) Not suitable for high current loads.

Applications:

-> low current DC power supplies
-> Electronic devices.



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